

DistalShift: How Distal Redundancy Shapes Robot Arm Motion and Joint Loading

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Abstract—Seven-degree-of-freedom manipulators provide redundancy for six-dimensional tool-pose control, yet local tool reorientation can still require substantial proximal-arm motion. We present a compact two-degree-of-freedom distal wrist that localizes tool reorientation near the endpoint. The Contact-Anchored Reorientation Envelope evaluates orientation coverage under a proximal-motion budget and supports comparisons of mounting and arrival strategies. Hardware experiments compare wrist-held and wrist-actuated configurations for ultrasound probe manipulation across five free-space and phantom-contact conditions, retaining the same installed hardware. Wrist activation reduced mean reconstructed incremental proximal swept volume by 44.5–95.8% and baseline-subtracted torque excursion by 18.9–89.4%. During axial probe rotation in phantom contact, peak and RMS measured arm-joint torque decreased by 48.4% and 29.8%, respectively. These gains were task dependent and accompanied by increased orientation-tracking error. Complementary diagnostic simulations indicate lower whole-system absolute joint mechanical work than direct probe mounting, including the wrist joints’ contribution, despite higher proximal torque RMS. Together, these findings characterize how distal participation localizes tool motion and provides task-dependent mechanical benefits.

Hardware designs and software are available at <https://distalshift.github.io/>.

I. INTRODUCTION

Seven-degree-of-freedom (DoF) manipulators provide redundancy for controlling six-dimensional tool poses, but endpoint capability alone does not characterize the motion required from the rest of the robot [1]. After a tool reaches its target, a local orientation adjustment can still require substantial shoulder, elbow, and forearm reconfiguration. Small tool reorientations producing large swept arm motions pose an acute risk for robots operating in medical environments, where robots perform tasks beside a patient, operator, or surrounding equipment. For such robots, an important design question is not only whether the tool can reach a pose, but how much proximal motion is needed to realize it.

Placing additional articulation near the tool offers a way to increase efficiency of motion. Macro–mini manipulation established the advantage of pairing a small distal mechanism

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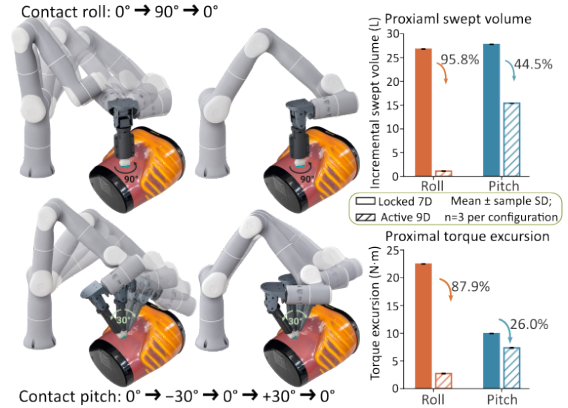


Fig. 1. Contact probe reorientation with a locked (7-DoF) or active (9-DoF) distal wrist. Left: simulation illustrations. Right: incremental proximal swept volume reconstructed from hardware trajectories and baseline-subtracted torque excursion (mean $\pm$ sample SD, $n = 3$ per condition).

with a larger positioning arm, including benefits in effective inertia and dynamic performance [2], [3]. A review of prosthetic and robotic wrists identifies compact packaging, low distal mass, and useful orientation range as recurring design considerations [4]. Recent work on compact independent-wrist architectures also emphasizes a complementary trade-off: serial wrists offer direct output joints and relatively simple kinematics, whereas parallel wrists can reduce moving mass at the cost of more involved kinematics and singularity analysis [5]. This distinction motivates treating wrist architecture and joint placement as design variables rather than counting added DoF alone. Prior work investigating the benefit of single-axis and 2-axis wrists for efficient operation in constrained spaces [6], [7] motivate studying not only the *number* of added joints, but their placement and use. Building on this foundation, we examine how equipping a commercial arm with a 2-DoF wrist changes both the spatial motion and measured joint loading during tool reorientation.

Robot-assisted ultrasound provides a representative setting for this question. Once the probe reaches an acoustic window, it must be tilted and rotated to refine a view or change imaging planes while maintaining acoustic contact. Measurements of sonographer motion have informed end-effector designs for local probe adjustment [8]. Larger orientation changes also occur: the transthoracic echocardiography guideline describes a 90° transducer rotation from the parasternal long-axis view to obtain a parasternal short axis view [9]. These maneuvers motivate evaluating small local tilts and larger axial rotations near a nominal contact location, rather than treating probe positioning as a sequence of unconstrained

reaches.

Robotic ultrasound systems have been developed for diagnostic and interventional tasks across multiple clinical settings [10]. Recent work has addressed complementary aspects of this interaction, including standardized acquisition with dual-probe systems [11], automatic surface-normal alignment [12], and end-effectors that provide adjustable constant contact forces [13]. Operational-space and hybrid position–force control offer established foundations for specifying tool-motion and contact objectives [14], [15]. Our focus is the whole arm cost of realizing those objectives: the proximal configuration changes, additional occupied space, and joint loads associated with a prescribed probe motion. Evaluating these quantities alongside pose and force tracking connects endpoint performance to the robot’s behavior in its surrounding workspace.

A smaller proximal motion envelope does not by itself establish lower measured arm loading. The arm still supports the installed wrist and tool, while configuration and contact-load transmission change during reorientation. We therefore examine when distal participation provides concurrent spatial and load benefits, and how these benefits relate to achieved task tracking.

We present DISTALSHIFT, an open-source 2-DoF wrist that provides local tilt and axial rotation between a commercial 7-DoF arm and its tool. A kinematic coordination layer combines wrist reorientation with the arm’s Cartesian motion–force controller and compensates for tool translation induced by wrist motion. Hardware trials compare wrist-held and wrist-actuated operation within the same installed assembly, evaluating proximal spatial sweep, measured joint loading, and achieved task tracking. The Contact-Anchored Reorientation Envelope (CARE) and configuration-envelope analyses complement these trials by characterizing mounting and arrival choices under a proximal-motion budget. A supplementary diagnostic simulation comparison adds direct probe mounting and evaluates whole-system absolute joint mechanical work across direct-probe, wrist-held, and wrist-actuated configurations, including the contribution of the distal joints.

Across five hardware task conditions, wrist activation reduced mean reconstructed incremental proximal swept volume by 44.5–95.8% and baseline-subtracted torque excursion by 18.9–89.4%. Contact roll additionally reduced peak measured proximal-joint torque by 48.4% and raw joint-and-time torque RMS by 29.8%. These results identify substantial spatial benefits and task-dependent load reductions, together with their tracking tradeoffs. Figure 1 highlights the contact-roll and contact-pitch comparisons.

The contributions are:

- An open-source 2-DoF distal wrist and an arm–wrist coordination architecture compatible with a commercial Cartesian motion–force controller.
- A contact-anchored kinematic evaluation of reorientation coverage under bounded proximal motion, including arrival-strategy and mounting comparisons.
- A hardware characterization of task-resolved proximal

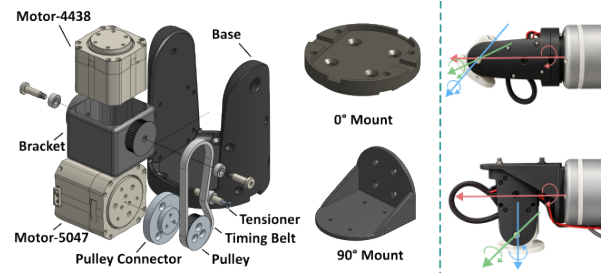


Fig. 2. 2-DoF distal wrist and mounting configurations. Left: exploded assembly with actuators and timing-belt transmission. Center: 0° and 90° mounting adapters. Right: corresponding assembled configurations.

spatial sweep, measured joint loading, and tracking under five conditions, complemented by a diagnostic comparison of direct-mount simulation of absolute joint mechanical work of the whole-system.

Although ultrasound motivates the evaluation, the formulation concerns tool reorientation near a fixed point or contact region and can guide similar studies in inspection and surface-interaction tasks.

II. SYSTEM AND CONTROL ARCHITECTURE

A. Open-Source 2-DoF Distal Wrist

The add-on wrist introduces two actuated rotational joints between the robot flange and tool without modifying the proximal manipulator. J8 provides local tilt, and J9 provides axial rotation aligned with the ultrasound probe’s acoustic axis. This distal arrangement places orientation motion near the tool [4], [16], while the arm supplies the remaining motion.

The mechanism comprises a mounting base, a pitch stage carrying the axial-rotation assembly, and a probe adapter (Fig. 2). J8 and J9 use HighTorque HTDW-5047-36-NE and HTDW-4438-32-NE planetary actuators with 36:1 and 32:1 gear ratios, respectively, and integrated encoders. The base and mounts are printed in Markforged Onyx; the bracket, pulley, and pulley connector use Formlabs Rigid 4000 Resin. The CAD assembly and URDF specify the mechanical layout and joint transforms.

The module weighs 720 g excluding the probe, with usable ranges of $\pm 90^\circ$ at J8 and $\pm 180^\circ$ at J9. The shortened probe adapter gives a nominal flange-to-contact-point distance of 290.4 mm. Interchangeable 0° and transverse 90° mounts are provided; all hardware experiments used the 90° assembly. Local wrist joint control operated at 500 Hz, while coordinated host references and experiment logging ran at approximately 100 Hz.

B. Arm–Wrist Kinematics and Coordination

Let $q_a = [q_1, \dots, q_7]^T$ and $q_w = [q_8, q_9]^T$ denote arm and wrist configurations. Frames W , F , C , and P denote the world, robot flange, configured fixed arm TCP, and probe frame at the nominal contact point, respectively. With ${}^A T_B$ expressing frame B in frame A , the probe pose is

$${}^W T_P(q_a, q_w) = {}^W T_F(q_a) {}^F T_P(q_w). \quad (1)$$

The rotation $R_w(q_w)$ and translation $r_w(q_w)$ of ${}^F T_P(q_w)$ follow the configured serial wrist geometry, including joint-origin and probe offsets. The commercial arm interface uses a fixed configured TCP, ${}^F T_C$, whereas wrist articulation changes the actual flange-to-probe transform. Scripted pitch θ and axial roll ϕ specify ${}^W R_{P,d} = {}^W R_{P,0} R_x(\theta) R_z(\phi)$, where ${}^W R_{P,0}$ is the initial probe orientation.

A kinematic coordination layer couples wrist actuation with the Flexiv non-real-time Cartesian motion–force interface, following the operational-space and hybrid position–force framework [14], [15]. For active operation, a bounded iterative orientation allocator produces q_w^{alloc} . Filtering, velocity and acceleration limits, joint margins, and a tracking-lead bound yield q_w^{cmd} ; successive commands determine wrist velocity references. The residual flange orientation and translation compensation are

$${}^W R_{F,d} = {}^W R_{P,d} R_w(q_w^{\text{alloc}})^T, \quad (2)$$

$${}^W p_{F,d} = {}^W p_{P,d} - {}^W R_{F,m} r_w(q_w^{\text{meas}}), \quad (3)$$

where d and m indicate desired and measured quantities.

Coordination therefore combines residual orientation assignment with offset compensation, using the allocated wrist configuration for orientation and measured wrist angles and flange orientation for translation. In locked operation, the wrist holds its starting angles with powered drives, and the flange target is ${}^W T_{F,d} = {}^W T_{P,d} [{}^F T_P(q_w^{\text{meas}})]^{-1}$. Both conditions use the fixed-TCP command ${}^W T_{C,d} = {}^W T_{F,d} {}^F T_C$.

In free space, the arm uses Cartesian impedance control and the wrist tracks position–velocity references. During contact, the arm regulates force along the z -axis of a world-fixed task frame aligned with the initial probe push direction n , while tangential translation and orientation remain motion controlled. The normal-position reference follows the measured probe position, allowing force regulation to adjust contact depth. Wrist tracking is supplemented by bounded rotational impedance assistance, gravity compensation, and contact-force feedforward. Orientation error and measured wrist angular velocity generate an impedance moment mapped to J8–J9 through the wrist Jacobian transpose. The host supervises contact engagement; continuous force regulation remains within the Flexiv controller and does not use the host force estimate as a replacement feedback signal.

C. Calibration and Measurement

Probe geometry uses tuned joint origins, axes, encoder offsets, and the probe-tip transform, with encoder zeros synchronized to the mechanical reference posture. The fixed Flexiv tool definition retains a nominal payload mass of 1.204 kg, distinct from the wrist-module mass. Host force observation subtracts an empirical unloaded-force model, affine in flange-frame gravity, and a session offset captured once before contact. The corrected force is rotated to the world frame and supplemented by a bounded nominal-payload-mass times acceleration term to obtain $\hat{f}_W$; acceleration is estimated from fixed-payload center-of-mass velocity and filtered. Positive compression is $f_c = -n^T \hat{f}_W$. Force-error evaluation uses this estimate before the final scalar

low-pass filter applied for contact supervision. The session offset remains fixed during contact; configurations outside the calibration domain use model extrapolation. Arm torque comes from the robot’s joint-torque channels. Probe pose and force metrics use kinematic reconstruction and processed flange sensing rather than independent external metrology.

III. TASK-DRIVEN KINEMATIC AND DYNAMIC ANALYSIS

A. Compared Simulation Strategies

The supplementary contact simulations compare direct-probe, wrist-held, and wrist-actuated configurations under shared probe-level references, timing, and contact conditions. Each configuration retains its respective mounting geometry and modeled distal mass. Whole-system absolute joint mechanical work is evaluated in Section III-F.

The offline kinematic study compares four allocation strategies: locked 7-DoF arm-only IK; generic 9-DoF IK with near-equal joint costs; distal-weighted 9-DoF IK that penalizes J1–J7 more than J8–J9; and a reorientation-aware method that selects the contact-arrival posture before using distal-weighted IK. The weighting is a null-space design choice, not a second controller or an equal-cost assumption [17].

The distal-weighted strategy uses

$$W_D = \text{diag}(25I_7, I_2). \quad (4)$$

Here, I_7 and I_2 correspond to the proximal arm and distal wrist. The 25:1 ratio favors J8 and J9 while allowing J1–J7 to move when required by pose feasibility, joint limits, or clearance.

B. Reorientation-aware arrival selection

At each contact, equal-weight nine-joint IK solves the same zero-tilt, zero-axial-rotation probe pose from seven deterministic seeds: the configured Home and six single-joint perturbations (J1: ± 0.28 rad; J3: ± 0.30 rad; J7: ± 0.45 rad). Converged solutions are deduplicated at a nine-joint Euclidean distance below 10^{-3} rad. Each candidate independently initializes distal-weighted IK for 15 coarse targets formed by tilt $\{-25, 0, 25\}^\circ$ and axial rotation $\{-90, -45, 0, 45, 90\}^\circ$, a subset of the full grid. Coarse success requires pose convergence within 2 mm and 2° , at most 8° J1–J7 RMS displacement from the candidate, and nonnegative target clearance. Candidates are ranked lexicographically by higher coarse success fraction, greater arrival clearance, and lower mean proximal RMS displacement over successful coarse targets; exact ties retain the first candidate. Arrival clearance is a ranking criterion, not a hard candidate filter. If all seven seeds fail, a Home-seeded fallback records arrival failure, and all full-grid targets remain infeasible. The selected arrival independently initializes all 143 targets using the unchanged target-level IK of the distal-weighted baseline, without inter-target warm starts or full-grid-based reselection.

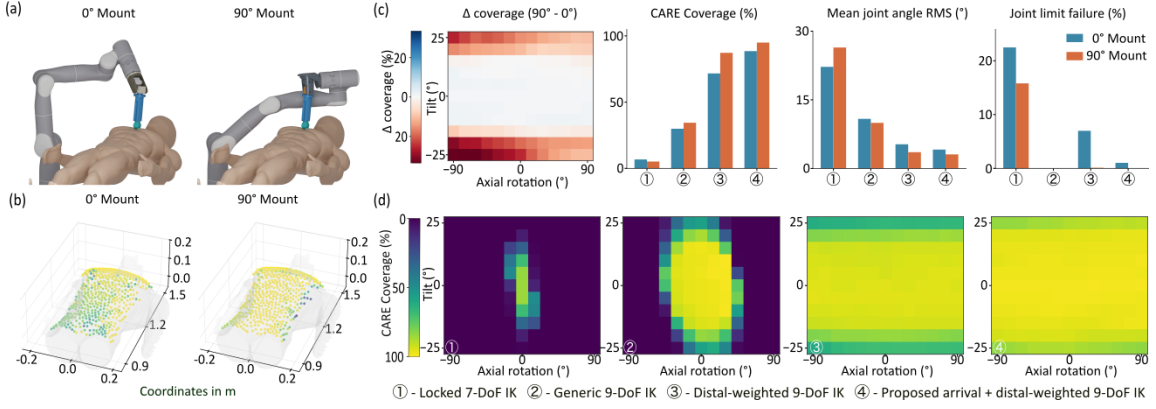


Fig. 3. CARE comparison of the original 0° and transverse 90° mounts. (a) Matched Isaac Sim renderings. (b) The 525 sampled torso contacts colored by proposed-method coverage over 143 tilt-axial-rotation targets. (c) Proposed-method coverage difference ($90^\circ - 0^\circ$), aggregate coverage, mean proximal J1-J7 RMS displacement, and joint-limit failure rate for the four methods. (d) Orientation-resolved coverage for the 90° mount. Heatmaps use a common 0–100% scale; positive difference means higher 90° coverage. Archived Home seeds and hardware revisions differ, so this is a complete-assembly comparison, not an isolated causal mount-angle ablation.

C. Clinically Motivated Reorientation Tasks

1) *Local Ultrasound View Adjustment*: The local-view task keeps the probe at the nominal parasternal contact region while commanding $0^\circ \rightarrow +5^\circ \rightarrow -5^\circ \rightarrow 0^\circ$ tilt at constant normal force in clinical [10], [8]. The $\pm 5^\circ$ motion represents fine view correction; optional $\pm 15^\circ$ and $\pm 30^\circ$ trials stress the extended range of a robotic ultrasound end effector. Both systems receive identical references.

2) *Imaging-Plane Switching*: The imaging-plane task rotates the probe about its acoustic axis as $0^\circ \rightarrow 90^\circ \rightarrow 0^\circ$ at the same contact point and normal-force command. This PLAX-to-PSAX-style transition follows the guideline description of an approximately 90° clockwise rotation while maintaining the parasternal window [9]; the return tests reversibility.

D. Reorientation Coverage Under a Proximal-Motion Budget (CARE)

The *Contact-Anchored Reorientation Envelope* (CARE) is the sampled set of probe orientations feasible at a fixed torso contact point under pose tolerances, joint limits, a geometric-clearance condition, and a J1-J7 motion budget. It is an offline capability-style kinematic benchmark [18], [19]; it does not measure force tracking, dynamic torque, tissue deformation, or continuous trajectory feasibility.

We sample $N_c = 525$ torso contact locations from the original patient mesh. At contact i , the neutral orientation is $R_{i,0}$, the acoustic-face center is p_i , and the target for local tilt α_g and axial rotation β_g is

$$R_{i,g}^d = R_{i,0} R_y(\alpha_g) R_z(\beta_g), \quad p_{i,g}^d = p_i. \quad (5)$$

Here, $R_y(\alpha_g)$ applies local tilt and $R_z(\beta_g)$ axial rotation about the tilted probe axis; $p_{i,g}^d = p_i$ keeps the acoustic-face center at the original contact point.

Tilt angles span -25° to $+25^\circ$ in 5° increments and axial angles span -90° to $+90^\circ$ in 15° increments, giving $N_g = 143$ targets per contact. The damped weighted IK

solver uses coefficient 0.008, at most 90 iterations, maximum joint increment 0.20 rad, and a 1° joint-limit margin.

For a fixed method m , contact i , and target g , let $\Delta q_{a,j}$ denote the change of proximal joint j between the neutral arrival configuration and the target IK solution. The RMS proximal joint displacement is

$$d_a = \frac{180}{\pi} \sqrt{\frac{1}{7} \sum_{j=1}^7 (\Delta q_{a,j})^2} [\text{deg}]. \quad (6)$$

This is the post-arrival RMS change of J1-J7 in degrees. J8 and J9 remain in the IK and joint-limit checks but are excluded from d_a , which measures only proximal-arm motion.

Let $\text{PF}_{i,g}^{(m)}$ denote the pose-feasibility indicator. It equals one only when the IK solution satisfies the translational tolerance of 2 mm, the rotational tolerance of 2° , and the 1° joint-limit margin; otherwise, it equals zero. For a proximal-motion budget B , the CARE success indicator is

$$I_{i,g}^{(m)}(B) = \text{PF}_{i,g}^{(m)} \mathbf{1}\{d_a \leq B\} \mathbf{1}\{c_{\text{proxy}} \geq 0\}. \quad (7)$$

Here, $\mathbf{1}\{\cdot\}$ is an indicator. The terms enforce pose feasibility, the J1-J7 RMS budget B , and nonnegative sampled clearance c_{proxy} . The proxy compares link-centerline envelopes with the torso height field and bed bounding box; it rejects sampled penetration but is not continuous mesh collision checking or a hardware-safety certificate.

CARE coverage is calculated over all sampled contacts and orientation targets, including failed targets in the denominator:

$$\text{CARE}^{(m)}(B) = \frac{1}{N_c N_g} \sum_{i=1}^{N_c} \sum_{g=1}^{N_g} I_{i,g}^{(m)}(B). \quad (8)$$

Thus, $\text{CARE}^{(m)}(B)$ is the fraction of all $525 \times 143 = 75,075$ targets satisfying pose feasibility, the proximal-motion budget, and sampled clearance. The primary analysis

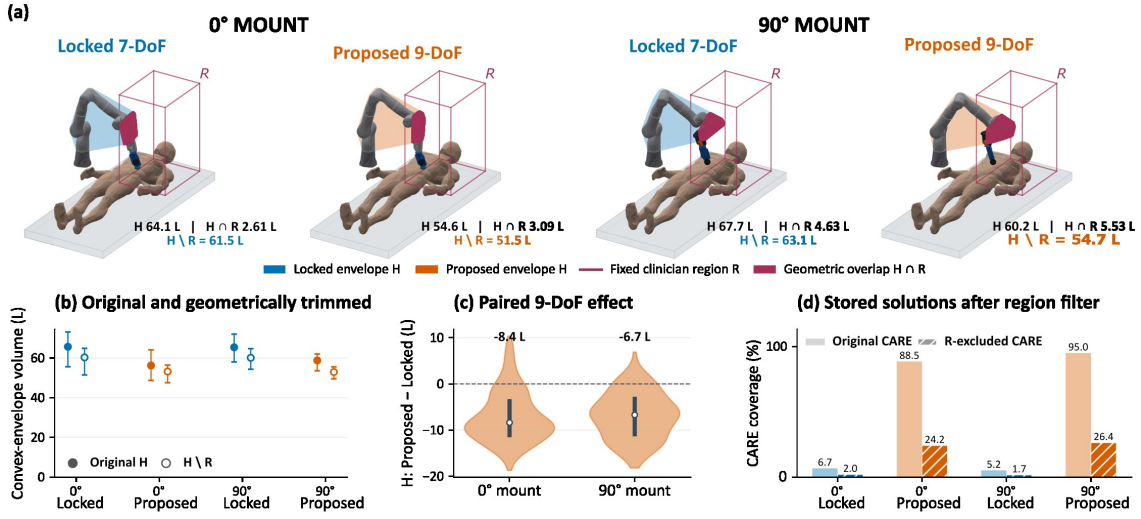


Fig. 4. CARE-based configuration envelopes and clinician-side screening for the 0° and 90° assemblies. (a) Proximal hulls H at a common representative contact, fixed nominal region R , and their geometric intersections. (b) H and $H \setminus R$ volumes; the latter is geometric trimming, not replanned motion. (c) Paired proposed-minus-locked hull-volume differences (median/IQR over 407 contacts with four-way shared targets). (d) Original and stored-target region-screened coverage, each using all 75,075 targets per condition. Envelopes include method-specific neutral arrivals; the main coverage screen evaluates stored target configurations.

uses $B = 8^\circ$ and compares locked 7-DoF, generic 9-DoF, distal-weighted 9-DoF, and reorientation-aware arrival strategies.

Because methods may use different neutral arrival postures, CARE compares complete system-and-arrival policies rather than a matched-start causal ablation.

Mounting and allocation results: For the 0° assembly, CARE coverage was 6.74%, 29.90%, 71.67%, and 88.50% for arm-only, generic 9-DoF, distal-weighted, and proposed arrival-aware strategies, respectively. For the 90° assembly, the corresponding values were 5.15%, 34.45%, 87.21%, and 94.96%. The intermediate nine-DoF comparisons show that coverage under the prescribed proximal-motion budget depends on allocation and arrival selection, rather than joint count alone. Within the evaluated mounting-and-arrival configurations, proposed-method coverage was 6.46 percentage points higher for the 90° assembly, providing a coverage-oriented mounting rationale.

E. Kinematic Envelopes and Clinician-Side Workspace

For each contact, we retain orientation targets successful in all four mounting-strategy conditions (0°/90° and locked/proposed). The hull H contains the transformed J1–J7 collision-mesh vertices at these stored solutions and each condition’s neutral arrival. This is a sampled configuration envelope, without trajectory interpolation. The numerical model retains the CARE pipeline’s J7 placement correction. Statistics include all 407 contacts with nonempty four-way intersections; empty sets are excluded rather than assigned zero volume. The representative contact in Fig. 4(a) is the centroid-nearest contact among those with at least five shared targets.

The fixed nominal clinician-side region is a box centered at (0.80, 0.06, 0.50) m with dimensions $0.50 \times 0.60 \times 0.90$ m, aligned with the common robot-base frame. Positive y points

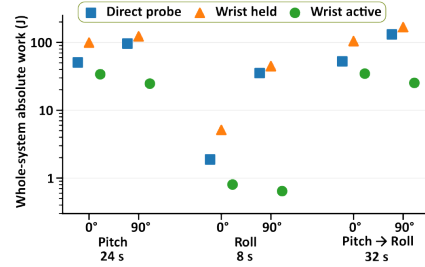


Fig. 5. Whole-system absolute joint mechanical work in 15 N contact simulations: pitch ($0^\circ \rightarrow +30^\circ \rightarrow -30^\circ \rightarrow 0^\circ$, 24 s), roll ($0^\circ \rightarrow 90^\circ$, 8 s), and their continuous sequence (32 s). The 0° and 90° groups retain distinct direct-probe baselines. Bars represent single runs; active totals include J8–J9. Results remain diagnostic: dynamics checks are unmet, with additional tracking violations in pitch and combined runs.

headward and positive x toward the clinician side. All methods and mountings use this same headward, robot-near layout. We report the geometric overlap $H \cap R$ and remainder $H \setminus R$. Separately, the stored-target screen rejects original CARE successes whose proximal mesh vertices enter R , without re-solving IK. Its denominator remains all 75,075 sampled targets; neutral-arrival intersections are logged separately and do not enter the main screen. This is a vertex-based layout screen rather than continuous collision checking.

On shared targets, the proposed strategy reduced hull volume relative to the locked baseline by a median paired 8.36 L (12.7%) at 0° and 6.71 L (10.3%) at 90°. Under the revised region screen, proposed coverage was 24.17% and 26.36%, compared with 2.03% and 1.65% for the corresponding locked baselines. Thus the proposed strategy combines smaller shared-target proximal envelopes with a larger set of accepted stored targets. The proposed 90° assembly retained 2.19 percentage points more coverage than its 0° counterpart, although its paired hull volume was 1.38 L larger at the median. Mounting selection therefore

TABLE I

SPATIAL AND MEASURED-LOAD STATISTICS (MEAN $\pm$ SAMPLE SD, $n = 3$ PER ROW). V_Δ IS RECONSTRUCTED OVER THE EXECUTED INTERVAL; P_τ , R_τ , AND E_τ USE MOTION SAMPLES ONLY.

Task	Wrist	V_Δ (L)	P_τ (N m)	R_τ (N m)	E_τ (N m)
Free axial	Locked	27.28 ± 0.03	53.10 ± 0.07	18.932 ± 0.005	23.226 ± 0.027
	Active	1.25 ± 0.01	30.24 ± 0.08	14.322 ± 0.015	2.454 ± 0.052
Free square	Locked	70.84 ± 0.04	65.89 ± 0.10	19.426 ± 0.009	21.816 ± 0.080
	Active	30.03 ± 0.09	56.22 ± 0.04	19.159 ± 0.022	11.691 ± 0.145
Contact roll	Locked	26.80 ± 0.05	48.94 ± 0.99	16.362 ± 0.012	22.491 ± 0.044
	Active	1.13 ± 0.01	25.27 ± 0.09	11.484 ± 0.007	2.729 ± 0.057
Contact pitch $\pm 15^\circ$	Locked	14.19 ± 0.02	57.63 ± 0.43	21.513 ± 0.007	5.490 ± 0.210
	Active	7.85 ± 0.01	55.54 ± 0.19	21.490 ± 0.011	4.450 ± 0.024
Contact pitch $\pm 30^\circ$	Locked	27.78 ± 0.04	60.88 ± 0.66	21.651 ± 0.006	9.948 ± 0.039
	Active [†]	15.41 ± 0.01	58.90 ± 0.31	21.781 ± 0.007	7.364 ± 0.108

balances reorientation coverage and spatial footprint rather than treating their benefits as interchangeable.

F. Whole-System Mechanical Work

To complement the same-assembly hardware comparison, we analyze 18 contact simulations spanning two mounting geometries and direct-probe, wrist-held, and wrist-actuated configurations, with shared nominal probe references and timing within each task. Modeled distal masses are 0.484 kg for direct mounting and 1.204 kg with the wrist. Whole-system absolute joint mechanical work is

$$W_{\text{abs}} = \int_{\mathcal{T}} \sum_{j \in \mathcal{J}} |\tau_{m,j}(t) \dot{q}_j(t)| dt, \quad (9)$$

using full gravity-inclusive simulated drive torque and actual joint velocity, without baseline subtraction. The sum includes J1–J7 for direct/held and J1–J9 for actuated operation. Across the six task–mount pairs, recorded W_{abs} was 33.1–98.2% lower in actuated operation than with the corresponding direct mounts (Fig. 5). The 90° actuated assembly also had lower absolute work than both evaluated direct mounts in every task. Proximal raw torque RMS was nevertheless 5.0–39.6% higher than the corresponding direct mounts. These diagnostic observations suggest reduced whole-system mechanical-work throughput, including the wrist’s contribution, rather than lower electrical consumption or uniformly lower joint loading.

IV. HARDWARE EXPERIMENTS AND RESULTS

Hardware experiments quantified proximal-arm spatial sweep and joint loading during scripted probe motion; pose tracking, processed force error, and execution pauses provide the associated task-fidelity context.

A. Experimental Protocol and Evaluation Metrics

a) Setup and comparisons: The setup comprised a Flexiv Rizon 4s arm, the 2-DoF wrist, a **Butterfly iQ+** probe [20], and an **ABDFAN** ultrasound-training phantom [21]. The contact protocol follows prior force-regulated

robotic-ultrasound end-effectors [13], [22]. Both configurations used the same 90° assembly, payload, scripted references, timing, and task-specific controllers (Section II-B). Initial states were nominally matched and recorded, with each trial’s initial arm configuration supplying the null-space reference. Locked J8–J9 used powered position holding with zero reference velocity. This same-assembly comparison evaluates the benefit of using the installed distal articulation together with its coordination policy. Arm translational stiffness was 3500 N/m per axis. Rotational stiffness and damping ratios were 675 N m/rad and 0.7 in free space, and 300 N m/rad and 0.8 in contact, where the normal direction was force controlled. Each of the five conditions in Table II comprised three trials per configuration (30 trials total). Conditions were analyzed separately, retaining records with tracking penalties or internal pauses.

TABLE II

SCRIPTED TASK CONDITIONS. EACH HAS THREE LOCKED AND THREE ACTIVE TRIALS. DURATIONS ARE NOMINAL MOTION TIMES, EXCLUDING APPROACH AND PRE-/POST-HOLDS.

Task	Probe reference	Time (s)	Force (N)
F-axial	$0^\circ \rightarrow 90^\circ \rightarrow 0^\circ$; 5 s hold at 90°	16	–
F-square	100 mm square; midpoint pitch $\pm 30^\circ$; corner roll $\pm 45^\circ$	416	–
C-roll	$0^\circ \rightarrow 90^\circ \rightarrow 0^\circ$	32	15
C-pitch	$0^\circ \rightarrow -\alpha \rightarrow 0^\circ \rightarrow +\alpha \rightarrow 0^\circ$; $\alpha = 15^\circ$ or 30°	32	15

F and C denote free-space and contact tasks. The two pitch amplitudes are distinct conditions. Active 30° contact pitch additionally incurred internal pauses, reported in the results.

b) Intervals and measured torque: The motion interval $\mathcal{M}$ contains samples labeled `motion_*`; preparation, approach, settling, holds, and pauses are excluded. For scalar x_k , timestamp-weighted RMS is

$$\mathcal{R}_{\mathcal{M}}(x) = \sqrt{\frac{\sum_{k \in \mathcal{M}} \Delta t_k x_k^2}{\sum_{k \in \mathcal{M}} \Delta t_k}}, \quad (10)$$

where Δt_k is the next timestamp difference and the last sample has zero weight. With τ_i the measured proximal torque and $\bar{\tau}_{i,0}$ its mean during the nominal 5 s initial hold,

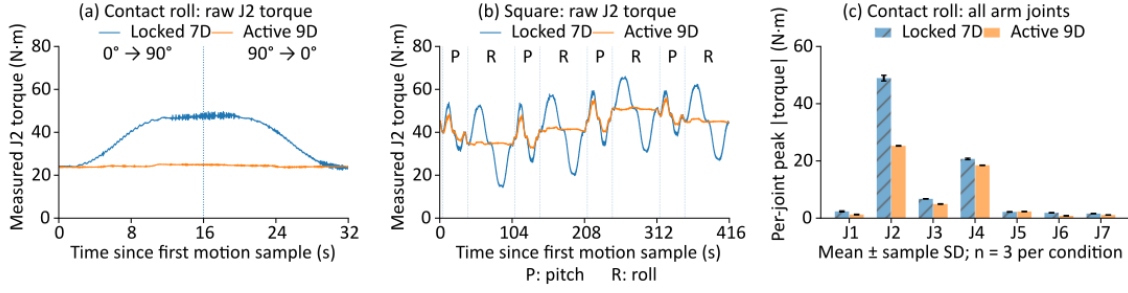


Fig. 6. Measured proximal torque. (a) Raw J2 contact-roll and (b) square traces for repeat 1 over the complete motion interval; no smoothing, warping, or fitted shifts. P/R mark pitch/roll and intervening segments are translations. (c) Per-joint contact-roll peak torque (mean $\pm$ sample SD, $n = 3$). J2 was the maximum-load joint and is not representative of all seven joints.

we report

$$P_\tau = \max_{i=1,\dots,7} |\tau_{i,k}|, \quad (11)$$

$$R_\tau = \sqrt{\frac{1}{7} \sum_{i=1}^7 \mathcal{R}_M(\tau_i)^2}, \quad (12)$$

$$E_\tau = \sum_{i=1}^7 \mathcal{R}_M(\tau_i - \bar{\tau}_{i,0}). \quad (13)$$

P_τ and R_τ summarize peak and raw RMS loading. The overview's complementary E_τ sums seven per-joint RMS values; it is not a resultant torque. Initial-hold subtraction does not remove changing gravity, contact, or friction loads, and none of these statistics is energy or an isolated inertial/velocity-product torque.

c) *Reconstructed spatial sweep*: Recorded J1–J7 trajectories drove the archived arm URDF and seven closed collision meshes. Let $O(t)$ be their occupied-solid union and O_0 the first-motion occupancy. Over the executed interval $\mathcal{T}_e$ (holds and pauses retained), we compute

$$U = O_0 \cup \bigcup_{t \in \mathcal{T}_e} O(t), \quad (14)$$

$$V_\Delta = \text{vol}(U \setminus O_0). \quad (15)$$

V_Δ excludes the initial region and counts overlaps and revisits once. The base, wrist, probe, cables, and phantom are excluded; this is a collision-model reconstruction, not direct volume metrology or a whole-system safety metric.

All trials used a world-aligned 2.5 mm voxel grid and a 0.625 mm sampled-vertex displacement threshold with interval boundaries preserved. Halving that threshold changed ten representative repeat-1 results by at most 1.36%; the four axial cases checked at 1.25 mm changed by at most 0.90%. These are sensitivity checks, not geometric error bounds.

d) *Aggregation and task fidelity*: Each trial contributes one value. Figures and load statistics report mean $\pm$ sample SD, with reductions from condition means; SD is within-setting scatter, not model uncertainty. CSV recomputation agreed with archived torque metrics. Table I summarizes spatial/load results; Table III retains motion-window pose/force RMSE and pauses internal to $\mathcal{T}_e$. Probe pose is arm–wrist FK and force is a processed flange-sensor estimate, not independent metrology.

B. Free-Space Reorientation and Composite Motion

a) *Axial reorientation*: The free-space axial task showed concurrent spatial and load benefits: wrist activation reduced incremental proximal swept volume by 95.4% and peak measured arm torque by 43.1%. Raw RMS torque also decreased, although active orientation-tracking error was larger.

b) *Composite square task*: Phase-resolved results show that reduced proximal motion was concentrated in reorientation. Summed J1–J7 travel was nearly unchanged during translation but decreased during pitch and especially roll, where it fell from 50.320 to 1.061 rad. The largest measured torque occurred during roll in locked operation and during pitch in active operation (Fig. 6(b)). Despite the smaller spatial sweep and torque excursion, motion-window raw RMS torque decreased by only 1.4%.

C. Phantom-Contact Reorientation

Contact tasks used a 15 N processed-compression setpoint, with contact depth governed by normal-force regulation during reorientation.

a) *Axial contact reorientation*: Contact roll combined a 95.8% reduction in incremental proximal swept volume with 48.4% lower peak measured arm torque and 29.8% lower raw joint-and-time torque RMS. J2 was the maximum-load joint (Fig. 6(a,c)). Processed normal-force RMSE was 0.176 N for locked and 0.126 N for active operation; active orientation and tangential tracking errors were larger (Table III).

b) *Contact pitch at two amplitudes*: Both pitch amplitudes retained spatial and torque-excursion benefits. Incremental swept volume decreased by approximately 45%, while excursion decreased by 18.9% at 15° and 26.0% at 30°. Raw RMS torque was nearly unchanged at 15° and increased by 0.60% at 30°. The active 30° trials included 2.43 ± 0.03 s of internal pauses and had larger processed force error (Table III). Pauses were retained in spatial reconstruction but excluded from the primary motion-only torque statistics; excursion remained lower when pauses were included.

These same-setting repetitions do not establish cross-workspace generalization, whole-system energy savings, or clinical imaging quality.

TABLE III

TASK-FIDELITY CONTEXT: MOTION-WINDOW MEANS OVER THREE TRIALS. THESE ARE NOT PEAK ERRORS OR INDEPENDENTLY MEASURED ABSOLUTE ACCURACIES; PAUSES ARE INTERNAL TO $\mathcal{T}_e$.

Task	Wrist	e_p (mm)	e_R (°)	e_f (N)	Pause (s)
F-axial	Locked	0.635	1.469	–	0.00
	Active	0.536	3.264	–	0.00
F-square	Locked	0.542	0.341	–	0.00
	Active	1.259	1.423	–	0.00
C-roll	Locked	0.354	0.657	0.176	0.00
	Active	0.611	1.614	0.126	0.00
C-pitch 15°	Locked	0.432	0.261	0.209	0.00
	Active	1.108	0.729	0.206	0.00
C-pitch 30°	Locked	0.614	0.476	0.605	0.00
	Active†	1.596	1.182	1.059	2.43

e_p : FK position RMSE in free space or tangential RMSE in contact; e_R : FK orientation RMSE; e_f : processed normal-force RMSE. The † condition contains internal pauses. Force estimation can extrapolate outside its calibration domain; small internal RMSE is not independent force-accuracy validation.

V. DISCUSSION

The CARE comparisons show that distal redundancy should be considered together with motion allocation and arrival selection. The proposed strategy increased sampled coverage and reduced shared-target proximal configuration envelopes relative to locked operation in both assemblies. Higher coverage for the 90° assembly, but not a uniformly smaller envelope, motivates evaluating mounting by both task access and proximal space.

The hardware results distinguish spatial savings from measured-load reductions. Axial reorientation provided both, whereas contact pitch retained smaller swept volumes and torque excursions with little change in raw torque RMS. A lower E_τ indicates smaller departures from the initial holding torque, not necessarily a lower raw torque magnitude. The square task reinforces this distinction: proximal travel decreased mainly during reorientation, while aggregate motion-window raw RMS changed little.

The direct-mount diagnostics further show that lower whole-system absolute joint mechanical work can coexist with higher proximal torque RMS, even when the wrist's own work is included.

Together, these findings motivate matching distal articulation and its coordination policy to the tool's dominant reorientation demands, rather than evaluating added DoF alone. The hardware benefits were obtained through the existing commercial motion-force interface. Increased orientation errors and the pauses in the largest pitch condition identify coordinated tracking as a priority for refinement; application suitability should be assessed against task-specific pose, force, and timing requirements.

VI. CONCLUSIONS

We presented an open-source two-DoF distal wrist integrated with a commercial seven-DoF arm. CARE and configuration-envelope analyses inform mounting and arrival choices. Across five hardware conditions, wrist activation reduced incremental proximal swept volume by 44.5–95.8% and baseline-subtracted torque excursion by 18.9–89.4%,

with additional raw-torque reductions in axial reorientation. Complementary diagnostic simulations indicate lower whole-system absolute joint mechanical work than direct probe mounting, including the wrist's own contribution, despite higher proximal torque RMS. Together, these results support task-aware mounting and coordination for localizing tool motion and obtaining task-dependent load and mechanical-work benefits, assessed alongside achieved tracking performance.

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